T13 $n_1 = 139$ $n_2 = 1000$

~~Нас~~: $\xi \sim N(a_\xi, \sigma_\xi^2)$

$\eta \sim N(a_\eta, \sigma_\eta^2)$

Длина:

$S_\xi^2 = 5{,}722$

$S_\eta^2 = 6{,}161$

Ширина:

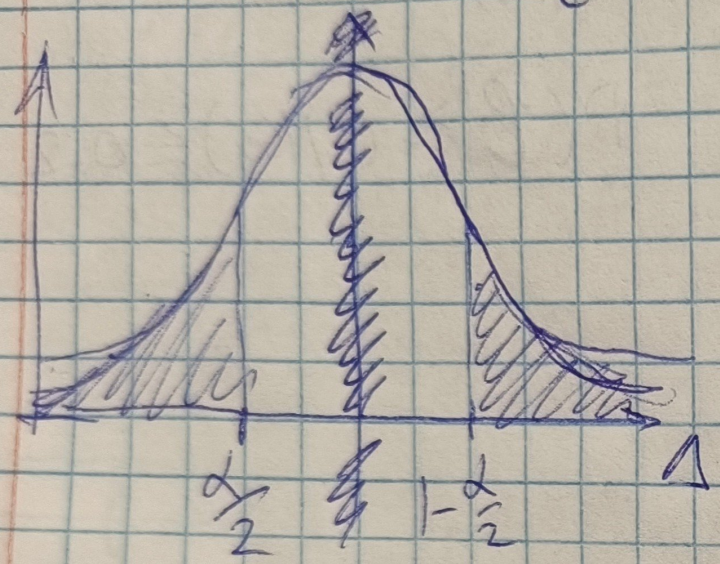
$S_\xi^2 = 4{,}612$

$S_\eta^2 = 5{,}055$

$H_0: \sigma_\xi = \sigma_\eta$, т.е. $\frac{S_\xi^2}{S_\eta^2} \sim F(n-1, m-1)$

$H_1: \frac{\sigma_\xi}{\sigma_\eta} = \vartheta \neq 1$, т.е. $\frac{S_\xi^2}{\vartheta^2 \cdot S_\eta^2} \sim F(n-1, m-1)$

Тогда $\Delta = \left(\frac{S_\xi}{S_\eta}\right)^2$



$G: \left[\begin{array}{l} \Delta \geq 1-\frac{\alpha}{2} \\ \Delta \leq \frac{\alpha}{2} \end{array}\right.$

Имеем:

$$\alpha_1 = P\left(\left[\begin{array}{l} \Delta \geq \tilde{q}_2 \\ \Delta \leq \tilde{q}_1 \end{array}\right. \Big| H_0\right) = \alpha$$

Т.е. $\Delta \sim a^2 F(n-1, m-1)$ – функция распр. Фишера

$$W = P\left(\left[\begin{array}{l} \Delta \geq 1-\frac{\alpha}{2} \\ \Delta \leq \frac{\alpha}{2} \end{array}\right. \Big| H_1\right) = \hat{f}\left(\frac{\alpha}{2a^2}\right) - 1 - \hat{f}\left(\frac{1-\frac{\alpha}{2}}{a^2}\right)$$

где $\hat{f}(x) = f(n, m, x)$ – плотность распр. Фишера

$$\frac{\alpha}{2a^2} = 0{,}767$$

$$\frac{1-\frac{\alpha}{2}}{a^2} = 1{,}272$$